National University of Singapore
School of Computing
CS3244: Machine Learning
Tutorial 09

Decision Trees and Ensemble Methods

1. **Decision Trees.**

   The loans department of DBN (Development Bank of NUS) has the following past loan processing records each containing an applicant's income, credit history, debt, and the final approval decision.

Income	Credit History	Debt	Decision
0 – 5K	Bad	Low	Reject
0 – 5K	Good	Low	Approve
0 – 5K	Unknown	High	Reject
0 – 5K	Unknown	Low	Approve
0 – 5K	Unknown	Low	Approve
0 – 5K	Unknown	Low	Reject
5 – 10K	Bad	High	Reject
5 – 10K	Good	High	Approve
5 – 10K	Unknown	High	Approve
5 – 10K	Unknown	Low	Approve
Over 10K	Bad	Low	Reject
Over 10K	Good	Low	Approve

   Useful Info:

   - $\log_2 \frac{x}{y} = \log_2$ x - $\log_2$ y,
   - $\log_2 1 = 0$
   - $\log_2 2 = 1$
   - $\log_2 3 = 1.585$
   - $\log_2 4 = 2$
   - $\log_2 5 = 2.322$
   - $\log_2 6 = 2.585$
   - $\log_2 7 = 2.807$
   - $\log_2 8 = 3$
   - $\log_2 9 = 3.170$
   - $\log_2 10 = 3.322$
   - $\log_2 11 = 3.459$
   - $\log_2 12 = 3.585$

(a) **Random Forest (Discussion).** Construct a random forest of size 3, where each of the three decision trees is fully grown from two of the three attributes, again based on the same set of examples: {Income, Credit History}, {Credit History, Debt} and {Debt, Income}.

2. **Analyzing Adaboost**

Algorithm 1 AdaBoost for Binary Classification

Initialize the data points weights to $\theta^{(1)} = [\frac{1}{m}, \frac{1}{m}, ..., \frac{1}{m}]$
for $t = 1, 2, ..., T$ **do**
  Fit a classifier $g_t(x)$ to the training data with weights $\theta^{(t)}$
  Compute the $\theta^{(t)}$- weighted 0/1 error ϵ_t by:

$$\epsilon_t = \frac{\sum_{i=1}^{m} \theta_i^{(t)} I(y_i \neq g_t(x_i))}{\sum_{i=1}^{m} \theta_i^{(t)}}$$

where $I(x)$ is a function that output 1 if x is true, 0 otherwise.
  Compute $\alpha_t = \frac{1}{2} \ln(\frac{1-\epsilon_t}{\epsilon_t})$
  Set $\theta_i^{(t+1)} \leftarrow \frac{\theta_i^{(t)} exp[-\alpha_t y_i g_t(x_i)]}{Z_t}$, $i = 1, 2, ..., m$ where Z_t is the normalizer in t-th iteration.
return $G(x) = sign[\sum_{t=1}^{T} \alpha_t g_t()x]$

From the above algorithm, we have:

$$\theta_i^{(t+1)} = \frac{\theta_i^{(t)} \exp\left[-\alpha_t y^{(i)} h_t\left(x^{(i)}\right)\right]}{Z_t}$$

Therefore, we have:

$$\begin{aligned}
\theta_i^{(T)} &= \theta_i^{(1)} \frac{\exp\left[-\alpha_1 y^{(i)} h_1\left(x^{(i)}\right)\right]}{Z_1} \times \cdots \times \frac{\exp\left[-\alpha_T y^{(i)} h_T\left(x^{(i)}\right)\right]}{Z_T} \\
&= \frac{1}{N} \frac{\exp\left[-y^{(i)} \sum_t \alpha_t h_t\left(x^{(i)}\right)\right]}{\prod_t Z_t}
\end{aligned}$$

From this we have:

$$\begin{aligned}
\text{Training Error}(H) &= \frac{1}{N} \sum_i I\left(y^{(i)} \neq H\left(x^{(i)}\right)\right) \text{ (Where } I \text{ is the Indicator Function.)} \\
&= \frac{1}{N} \sum_i I\left(y^{(i)} f\left(x^{(i)}\right) \leq 0\right) \\
&\leq \frac{1}{N} \sum_i \exp\left(-y^{(i)} f\left(x^{(i)}\right)\right) \\
&\text{Since } e^{-z} \geq 1 \text{ If } z \leq 0 \\
&= \sum_i \theta_i^{(T)} \times \prod_t Z_t \\
&= \prod_t Z_t (\text{Using } *)
\end{aligned}$$

Note that $\sum_i \theta_i^{(T)} = 1$ since θ_T is a distribution

So we have defined an **upper bound** for the training error. However, it it not clear yet whether this upper bound decreases over multiple iterations and converges to a limit.

Actually we can further derive the upper bound of the training error that depends on ϵ_t(the missclassification rate of the tth classifier; ranging from 0 to 1).

Below, we given you three options. **Which is the best upper bound for the training error?** (one of them is not a correct upper bound, one of them is a looser upper bound, and the final one is the best upper bound)

A. $\exp\left(-2\sum_{t=1}^{T}\gamma_t^1\right)$

B. $\exp\left(-2\sum_{t=1}^{T}\gamma_t^2\right)$

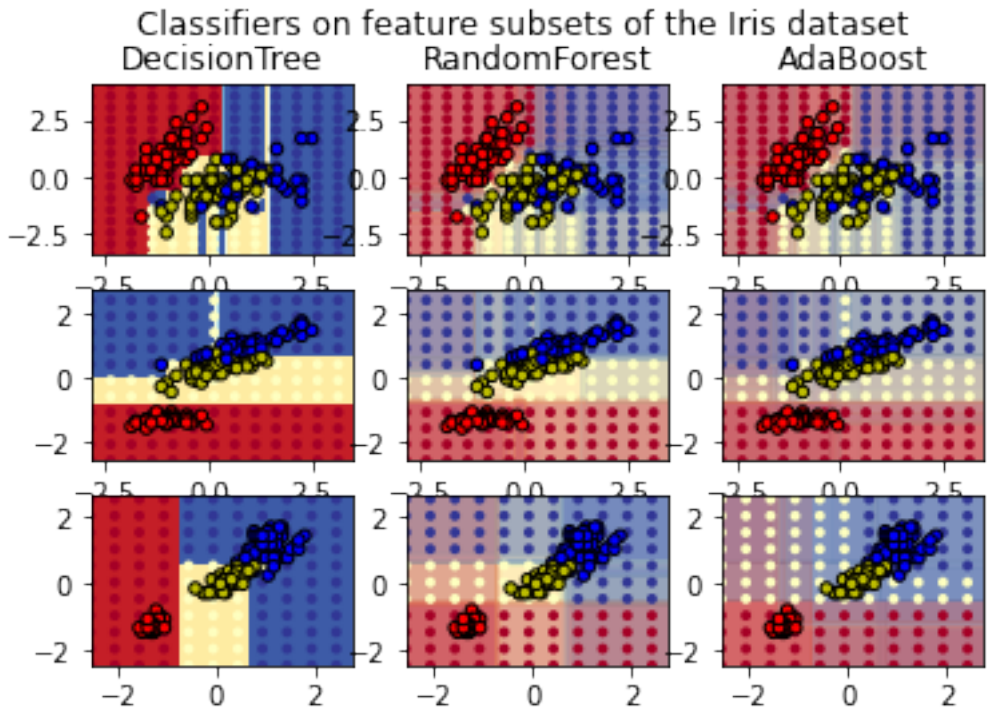
C. $\exp\left(-2\sum_{t=1}^{T}\gamma_t^3\right)$

HINT: Calculate Z_t, where $Z_t = \sum_i \theta_i^{(t)} \exp\left[-\alpha_t y^{(i)} h_t\left(x^{(i)}\right)\right]$

3. **Visualizing Decision Tree Boundaries**

Let's compare the decision boundaries of Decision Trees, Random Forests and the AdaBoost classifiers with Decision Stumps to classify on the Iris dataset.

For the sake of visualization, we will choose just two features from the set of 5 features of Iris dataset. Each experiment will make use of exactly two features. We will experiment with these 3 pairs of features: 0 and 1, 1 and 2, 0 and 2.



(a) From the visualization above, is Random Forest doing better than Decision Tree? Will the result still be the same, or improve if we have less estimators (say 10) for Random Forest?

(b) The vizualization used a seed of 13. What if the seed was changed to 119? Will the performance of three classifiers, decision tree, random forest and adaboost will still be same? If not, why?

4. **Uniform Blending(Optional)***.** One of the simplest ensemble methods is Uniform Blending. Given a set of hypothesis: $h_1, h_2, h_3, ..., h_T$, Uniform blending makes predictions simply by mixing the predictions given by $h_1, h_2, h_3, ..., h_T$ uniformly. Concretely, for binary classification, uniform blending predicts by:

$$H(x) = \text{sign}(\sum_{t=1}^{T} h_t(x)), \tag{1}$$

and for regression, uniform blending predicts by:

$$H(x) = \frac{1}{T}\sum_{t=1}^{T} h_t(x) \tag{2}$$

Taking regression as an example, show that the performance (measured by out-of-sample error) of uniform blending is no worse than the average performance over $h_1, h_2, h_3, ..., h_T$ i.e.

$$\frac{1}{T}\sum_{t=1}^{T} J_{\text{test}}(h_t(x)) \geq J_{\text{test}}(H(x)) \tag{3}$$

Assume we evaluate the testing error by mean square error.

Hint: Start by calculating the average error over $h_1, h_2, h_3, ..., h_T$ for one fixed data point x.

Proving equation 3 can give us an intuition on why ensembling can help to reduce the out-of-sample error.